

Pavel Košan

Data Engineer and Analyst

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About me

Since my bachelor's degree, I have been systematically involved in the field of data management - data storage, management, processing, modeling, and advanced analysis - with a particular interest in machine learning. I have applied the theoretical knowledge I have acquired in several real-world projects in which I have participated during my academic and professional practice.

Work experience

July 2023 – still
Pilsen, Czechia

Data Engineer and Analyst

Digitální architekti s. r. o.

- Building, orchestrating and maintaining ETL/ELT pipelines
- Optimizing BigQuery calculations and reducing analytics costs
- Customer analytics: RFM and cohort analysis, customer segmentation, customer 360
- Data mining and machine learning: MBA, clustering, recommendation models
- Developing python components for data pipelines and/or process automation
- Data preparation for conversational AI
- Tech stack: GCP (BigQuery, Cloud Run, Cloud Storage, Vertex AI, Dataform, Firebase, etc.), dbt, Keboola, Python, JavaScript, SQL

September 2025 – January
2026
Pilsen, Czechia

University lecturer

Department of Computer Science and Engineering, Faculty of Applied Sciences UWB

- Leading two groups of 20 students focusing on data processing, modeling and visualization
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Education

M. Sc. (Ing.): Software and Information Systems,

September 2023 – June 2025 | Pilsen, Czechia

Focus: Intelligent Information Systems

Faculty of Applied Sciences UWB

- Thesis: Possibilities of Using Multimodal Vector Data Representations in Personalized Medicine
- State Exams: **Data Analysis and Machine Learning** (Advanced Statistics, Machine Learning Algorithms, Visualization), **Data Processing Systems** (Databases, DWHs, Big Data, NLP, IR systems, Neural IR, Semantic Search, LLMs a RAG), **Software Systems Development** (SW Engineering, System Integration, Project Management)

B. SC. (Bc.): Computer Science and Engineering

September 2020 – June 2023 | Pilsen, Czechia

Faculty of Applied Sciences UWB

- Thesis: The Use of Deep Learning in Medical Applications
- PIT Award for the Best Bachelor Thesis
- State Exams: Data Structures and Algorithms, OS Fundamentals, IS Fundamentals

Courses and certificates

AI SUMMER SCHOOL

September 2025

Speinshart Science Center for AI and SuperTech, Germany

- Theme: **AI and Robotics**
- Practical programming of robotics arm Niryo Ned2
- Week hackathon
- Professional lectures

September 2023

AI SUMMER SCHOOL

Speinshart Science Center for AI and SuperTech, Germany

- Theme: **AI and Data Science**
 - Advanced statistical methods in Data Science
 - Data analysis and visualization in Python
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Tools and Technologies

Scale: basics, intermediate, advanced, very advanced

Programming Languages

SQL – very advanced
Python - advanced
Java, C, R, SPARQL, MQL – basics

Databases and Data Warehouses

BigQuery – very advanced
Snowflake, MySQL – intermediate
PostgreSQL, MongoDB, Neo4J - basics

Cloud Platforms

Google Cloud Platform (GCP) – intermediate

Machine Learning and AI

Scikit-learn, PyTorch – advanced
Keras, BigQuery ML, Vertex AI – intermediate

Visualization Tools

Looker Studio – very advanced
PowerBI, plotly dash, streamlit - intermediate

Other Data and Analytics Tools

dbt, Keboola – very advanced
DAX - advanced
Statistica – basics

Other Technologies and Tools

VSCode, GitHub, Postman - advanced